

6 Arithmetic and Logic Instructions

- Unpacked = 4 0s out front
- Packed = no 0s out front, more efficient.
 - However, there are reasons for having 0000s out front
 - Could process nibble, instead of byte by byte
 - Makes processing easier
- BCD addition **does not** work.. kind of.
- DA takes into account the AC as well.
 - If AC = 1 or > 9, add 06H
 - If CY = 1 or > 9, add 60H.
- The extra B in SUBB mean subtracted with borrow.
 - Make sure to set clear CY before using SUBB
- One special case, divide by zero: OV = 1, values remain the same.
 - Division example: A = 9, B = 5
- XRL only works for 8 bits.
 - Same addressing modes as for ANL
- Complement works for A, C or *anything* that is bit addressable.
- CJNE changes the CY flag
- Serial Communication example (we use RLC because we want to use the Carry flag for transmitting data)

ORG 0

MOV A, #35H

MOV P2, #0

MOV R0, #8

```
SETB P2.1
SETB P2.1
```

```
TX: RLC A
    MOV P2.1, C
    DJNZ R0, TX
```

```
SETB P2.1
SETB P2.1
```

```
END
```

- Same example, backwards

```
ORG 0
```

```
MOV R0, #8
```

```
RX: MOV C, P2.5
    RRC A
    DJNZ R0, RX
```

```
MOV R2, A
```

```
END
```

Number of '1's example

```
ORG 0
MOV R0, #0 ; Counter for 1s
MOV R1, #8 ; Counter for loop
```

```
MOV A, P2
```

```
LP: RRC A
```

```
DJNE
```

$$n\text{bit} \cdot n\text{bit} = 2n\text{bit} \quad (1)$$

```

ORG 0
MOV R0, #30H
MOV @R0, #0

XCHD A, @R0      ; M[30] = 07, A = 30H
SWAP A           ; A = 30H
ORL A, #30H

```